

Plan cutting, machining and assembly

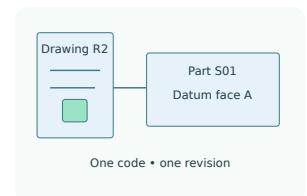
A production plan connects each part to a drawing revision, operation sequence and inspection point. Learn to release work in controlled stages instead of treating a cutting list as a complete manufacturing instruction.

What you will accomplish

- Freeze the production inputs
- Order operations by dependency
- Place a check before repetition

Freeze the production inputs

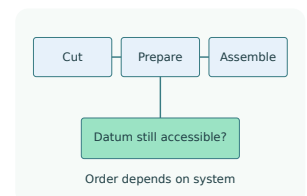
Start with an approved drawing revision, stable part codes, material, thickness, grain direction and exposed edges. Distinguish finished size from any process allowance defined by the production method. If a drawing changes, identify affected parts and withdraw obsolete instructions. A part should point to one controlling revision, not leave conflicting dimensions for the operator to resolve.



Part codes connect drawing, parts list and operation sheet.

Order operations by dependency

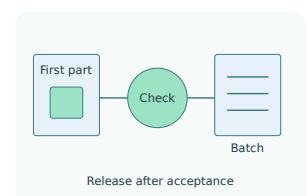
For a teaching cabinet side, identify a datum face, cutting, the system-specific order of edge treatment and holes, then trial assembly. Ask whether the next operation will conceal a measurement surface or block tool access. Some faces may need finishing before assembly. There is no universal sequence; this plan describes outputs and checks, not machine operating procedures.



Concealed references and access affect the operation order.

Place a check before repetition

Before repeating the operation across a batch, check the first part against the drawing: code, datum, finished dimensions and prepared features. Record the result, reviewer and release decision. If a mismatch appears, isolate potentially affected parts and identify the last verified stage. Forcing parts together does not replace dimensional checking or correction of the cause.



First-part approval precedes batch release.

Worked case and decision

A teaching batch contains 12 sides. Inspection of the first finds a hole located from the wrong datum. Hold the batch, correct the reference and instructions, and decide the disposition of that part before producing the remaining 11. A tight schedule does not justify repeating the mismatch.

Decision: hold repetition, correct the datum, then recheck the first conforming part.

Check understanding

When should the operation be released across the batch?

- After first-part inspection and documented acceptance
- After producing any first part
- When setup time expires

Correct: After first-part inspection and documented acceptance. Early verification prevents a reference or dimension error from propagating.

Lesson questions

Does an operation sheet qualify someone to operate a machine?

No. Machine operation requires separate competent training, approved instructions and appropriate supervision.

Practice

Prepare a cabinet-side operation sheet with part code, revision, three dependent stages, inspection point and mismatch response.

Part and controlling revision

Sequence and outputs

Inspection and release decision

Review your work against the criteria. Saving it does not constitute technical project approval.

- One dimensional reference
- Reasoned dependencies
- No invented machine settings